Water Resources Project Summary

Streamflow forecasting, satellite observation methods and a river-basin digital-twin prototype

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Project: Hydrology Institute project

Current status: The streamflow benchmark is implemented and has recorded results. The digital-twin and satellite modules are implemented as a research prototype. The allocation replay is exploratory; satellite processing in Earth Engine and independent field validation remain pending.

Project objective

Develop a reproducible workflow that predicts next-day river discharge, represents forecast uncertainty, and connects available environmental observations to transparent water-allocation scenarios. The project also documents the suitability and availability of candidate constructed-wetland phosphorus datasets.

Work completed

The repository contains four neural forecasting architectures, persistence and equal-average baselines, calibrated forecast Bayesian model averaging (BMA), saved predictions and evaluation metrics. Extensions cover rice classification, surface wetness, flood and slope-change screening, turbidity indices, observation availability, demand scenarios, constrained allocation, water-balance accounting, and a replay of saved forecasts.

Principal findings

For 2,192 test days at the Potomac River station, stacked LSTM achieved the lowest RMSE (111.22 m³/s), while BMA achieved the lowest MAE (37.79 m³/s). The allocation replay showed that conservative supply planning reduced overcommitment but increased unmet hypothetical demand. No real satellite-derived performance or operational water savings are claimed.

Repository: github.com/doraem-on/Hydrology_Institute-project
Reviewed source revision: bcf55f8

Report guide: data (p. 2); forecasting methods and results (pp. 3-4); satellite modules (p. 5); digital twin (p. 6); allocation replay (p. 7); implementation and verification (p. 8); limitations and next steps (p. 9); evidence and references (p. 10).

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Datasets and experimental scope

The project has three distinct data streams: an executed daily streamflow benchmark, an audit of wetland phosphorus sources, and a satellite-observation workflow awaiting study-specific inputs. These streams are not interchangeable.

Source	Recorded status and role
USGS 01646500	Downloaded and preprocessed. Potomac River near Washington, DC, Little Falls Pump Station; daily mean discharge, 1990-2025. Used in forecasting and replay.
Webb et al. (2025)	Supplementary ZIP and Table S2 workbook acquired. Study-level constructed-wetland phosphorus information; not used as a daily flow sequence.
Anderson et al. (2024)	Publisher supplements and associated Old Woman Creek source identified; access blocked during the recorded acquisition audit. Raw files not obtained.
UEA / Cooper (2020)	Ingol and Mun wetland water-quality catalogue identified; CSV download unsuccessful (HTTP 403). Sparse monthly sampling is unsuitable for this daily benchmark.
Dykes et al. (2025)	Reference paper obtained locally; underlying observations available on request and not acquired. The paper is not a raw dataset.
Satellite and field inputs	Sentinel-1, Sentinel-2, MODIS, terrain and surface-water products specified. Verified basin boundary, crop labels and paired field measurements remain required.

USGS quality control

The archive has 13,149 daily observations from 1 January 1990 to 31 December 2025. All have approved status; 174 also have an estimated-value flag, which is retained. Discharge is converted from ft³/s to m³/s using 0.028316846592. Processing rejects unapproved, negative, nonfinite and sentinel values and duplicate dates. Windows crossing missing dates or values are skipped rather than interpolated.

Partition	Dates of forecast targets	Samples	Purpose
Training	31 Jan 1990 - 31 Dec 2009	7,275	Scaling and model fitting
Validation	2010-2014	1,826	Epoch selection / early stopping
Calibration	2015-2019	1,826	Point-forecast correction and BMA
Test	2020-2025	2,192	Final recorded comparison

Acquisition status above reflects the project record, principally dated 8 October 2026; access was not rechecked for this report. Checksums and attribution are retained in the data manifest. The Potomac is the flow benchmark, not an established paddy-mapping case study.

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Forecasting methods

Shared prediction task

Each model predicts the next day’s mean discharge from the preceding 30 days. Inputs contain standardized log1p discharge and annual sine/cosine calendar features. Scaling uses training data only. The neural output is a residual correction to the final standardized log-flow value in the input window.

Method	Implementation and purpose
Persistence	Predict the next day using the last observed flow. Provides the minimum-complexity reference.
LSTM	One recurrent layer, 32 hidden units, and a linear prediction head. Gated memory summarizes the observed sequence.
Stacked LSTM	Two recurrent layers of width 32, with inter-layer dropout. The second layer models patterns in the first layer’s outputs.
Bidirectional LSTM	Forward and reverse encoders read the same past-only window. Both terminal hidden states are combined; no target-day or future flow is supplied.
Transformer	Two encoder layers, four attention heads, width 32, sinusoidal positions, and the final token representation. Attention covers historical inputs only.
Equal average	Arithmetic mean of the four corrected neural point forecasts. Tests whether a fitted ensemble adds value over simple averaging.
Forecast BMA	A calibrated mixture of predictive distributions built from the four neural models. Produces a nonnegative mean forecast and prediction intervals.

Training and forecast protocol

The recorded CPU run used seed 42, Adam with learning rate 0.001, batch size 128, a maximum of 20 epochs, and early-stopping patience of five. The training loss is mean squared error in standardized log1p flow. Calibration-period smearing factors correct the transformation to point forecasts in discharge units. Width is shared, but parameter counts differ.

How BMA represents uncertainty

For $z = \log(1 + Q)$, the fitted density is a weighted sum of Normal distributions with component mean $a_k + b_k f_k$ and a shared variance. Least squares fits the component corrections on the calibration period; expectation-maximization fits nonnegative weights summing to one and the residual variance. The implementation transforms the mixture using $Q = \max(\exp(z) - 1, 0)$, calculates its mean, and inverts the mixture CDF for interval bounds.

This is empirical forecast BMA, not a full Bayesian posterior over neural-network parameters. The test is a rolling next-day hindcast: earlier observed test flows may become inputs on later days. It is not a six-year forecast issued before 2020.

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Recorded forecasting results

The following scores use the same 2,192 daily test targets in 2020-2025. RMSE and MAE are in m³/s; lower is better. NSE and KGE are dimensionless, with values closer to one preferred. These are archived results from the 8 October 2026 run.

Model	RMSE	MAE	NSE	KGE (2009)
Persistence	145.79	48.54	0.7619	0.8809
LSTM	121.31	40.71	0.8351	0.9128
Stacked LSTM	111.22	37.89	0.8614	0.9143
Bidirectional LSTM	121.57	40.29	0.8344	0.9143
Transformer	116.79	40.32	0.8472	0.9031
Equal average	115.14	39.44	0.8515	0.9197
Forecast BMA	111.87	37.79	0.8598	0.9160

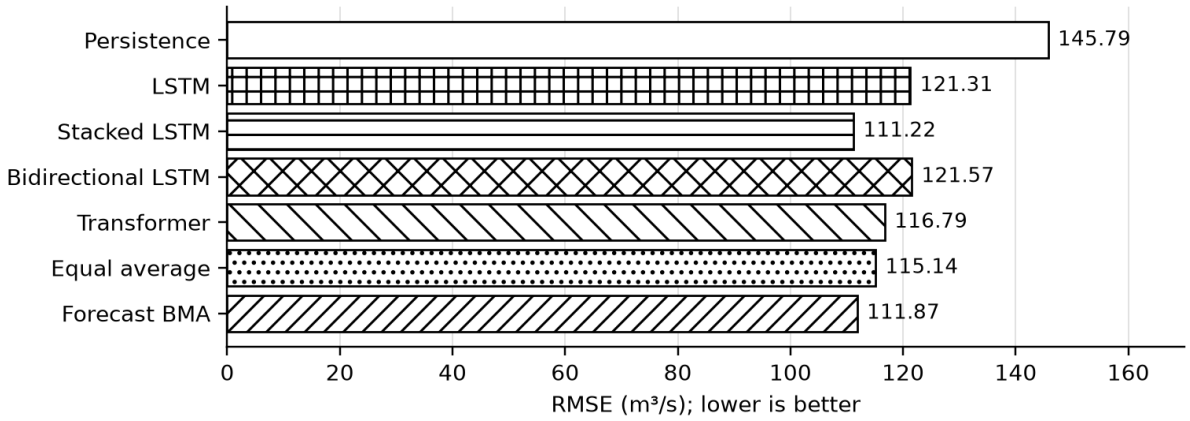


Figure 1. Test RMSE for all seven forecasting approaches. Values are reproduced from the saved metrics; hatching distinguishes bars in black-and-white printing.

Interpretation

Stacked LSTM has the lowest RMSE and highest NSE; its RMSE is 23.7% below persistence on this split. BMA has the lowest MAE. Equal averaging has the highest KGE. The comparison therefore does not identify one model that dominates every criterion.

BMA uncertainty and component weights

The nominal 90% prediction interval covered 92.97% of test observations; mean width was 156.39 m³/s. Fitted weights were LSTM 0.027%, stacked LSTM 76.706%, bidirectional LSTM 6.695%, and Transformer 16.572%. EM converged after 1,815 iterations.

Coverage is an empirical test-period result, not a guarantee for future conditions. The near-zero LSTM weight reflects calibration-period mixture support, not general uselessness of the architecture.

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Satellite observation modules

The satellite extension implements five established observation methods. Array kernels and CSV pipelines are tested locally; Earth Engine exports have not been executed and no site-specific mapping accuracy is reported.

1. Rice / paddy classification

Sentinel-1 VV and VH time sequences supply monthly features, temporal summary statistics, and VV - VH contrast in dB. A random forest uses 200 trees, minimum leaf size three, balanced class weights and seed 42. Median imputation is fitted on training data only. Training and test spatial groups must be disjoint. Outputs include rice probabilities, F1, balanced accuracy, average precision, Brier score and a confusion matrix. The current pipeline is retrospective end-of-season classification; independent labels are needed for field accuracy.

2. MODIS surface wetness (TOTRAM)

MOD13A2 NDVI and MOD11A2 land-surface temperature are quality-screened and scaled. Complete 8-day temperature composites are matched inside a 16-day vegetation window at approximately 1 km support. Conditional 5th/95th temperature percentiles in NDVI bins define fitted wet/dry edges, initially stratified in 200 m elevation bands. The index is $W = (T_{dry} - T)/(T_{dry} - T_{wet})$. Sparse, narrow-range or crossing-edge fits are rejected. Clipped and unclipped outputs are retained. This is relative surface wetness, not volumetric or root-zone moisture.

3. Flood candidate screening

Before/after Sentinel-1 imagery uses a common orbit, pass, polarization and grid. Initial controls are a 3 dB backscatter drop, post-event backscatter at most -15 dB, slope at most 5°, and at least eight connected pixels; persistent water is masked. Thresholds require event-specific validation. Vegetated and urban flooding may be missed.

4. Turbidity index

Cloud-screened Sentinel-2 water pixels produce NDTI = (red - green)/(red + green). Invalid or non-water pixels remain missing. NDTI is a dimensionless proxy; conversion to NTU or suspended-solids concentration requires paired measurements and calibration. It is not a water-safety classification.

5. Landslide / slope-change candidates

A bidirectional absolute SAR amplitude change of at least 4 dB on slopes of at least 15°, followed by component filtering, produces a review layer. The method does not establish landslide occurrence, displacement, coherence or future hazard. Terrain shadow, layover and other causes of change need independent assessment.

Earth Engine workflow

The two scripts export field-level SAR and MODIS sample tables plus event-screening rasters. They require a verified region, field labels and spatial blocks, suitable dates, a selected orbit and a local export CRS. Scene identifiers and observation-support metadata are recorded; historical availability is explicitly unverified. CSV commands support moisture fitting and rice evaluation while retaining provenance and rejected-scene records.

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Digital-twin state and decision logic

The prototype links observations, forecasts and allocation rules through an explicit observation-to-decision workflow. It is an experimental software framework, rather than a live synchronized basin-management system.

Stage	Implemented behavior
Observation record	unit_id, variable, observed_at, available_at, value and source_id. observed_at is the end of the observation interval.
Availability filter	Use records released by the issue time, whose observation interval has ended, and whose age is within the configured maximum (default 32 days). Reject ambiguous duplicate vintages.
Management-unit state	Keep the latest eligible crop fraction and surface-wetness observation for each unit. Preserve source and timing metadata.
Abstention / review	Missing or stale crop/wetness inputs stop allocation rather than implying zero demand. Optional flood and slope-change fractions generate review flags.
Demand and supply	Convert user-specified area, crop mix and wetness response to hypothetical demand. Use forecast supply and explicit reserve/diversion controls.
Allocation and audit	Apply proportional shortage, return unit allocations and unmet demand, and retain observation provenance. A separate bucket tracks mass balance.

Demand scenario

For unit area A in hectares, rice fraction p, wetness W, demand depths d_r and d_o in millimetres, response coefficient α and delivery efficiency η:

$$D = (10A/\eta) \times [p\, d_r + (1 - p) d_o] \times (1 - \alpha W).$$

The factor 10 converts hectare-millimetres to cubic metres. Crop-water requirements and the wetness response are scenario parameters requiring independent calibration.

Constrained allocation

A supply budget uses a selected forecast quantile q_τ, environmental reserve q_e, diversion fraction f and decision interval Δt:

$$B = f \times \Delta t \times \max(q_\tau - q_e, 0).$$

Each unit receives its demand multiplied by min(1, B / total demand). The reserve is applied before diversion. This is a transparent proportional rule, not an optimization of water rights, crop yield or economic welfare.

Physical accounting

The separate single-bucket model enforces S_{next} = S + rainfall + irrigation - runoff - actual ET. Storage is capacity-bounded and actual evapotranspiration cannot exceed available water. Mass-balance residuals are returned for checking. Routing, groundwater, reservoir operation, return flows and paddy ponding are not included.

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Exploratory water-allocation replay

Saved Potomac forecasts were replayed over the same 2,192 test days without retraining. Three planning inputs were compared: persistence, BMA mean, and the BMA 5th percentile (the lower endpoint of its 90% interval).

Scenario assumptions: Three hypothetical units each demand 1 million m³/day, for 3 million m³/day in total. The environmental reserve is 20 m³/s and the diversion fraction is 10% of flow above the reserve. These are not calibrated Potomac requirements or legal limits. No satellite observations enter this replay.

Planning input	Overcommitment days (%)	Mean unmet demand (million m³/day)	Mean overcommitment (m³/day)
Persistence	54.47	1.5893	63,317
BMA mean	63.55	1.5688	78,526
BMA 5th percentile	1.46	1.8381	2,717

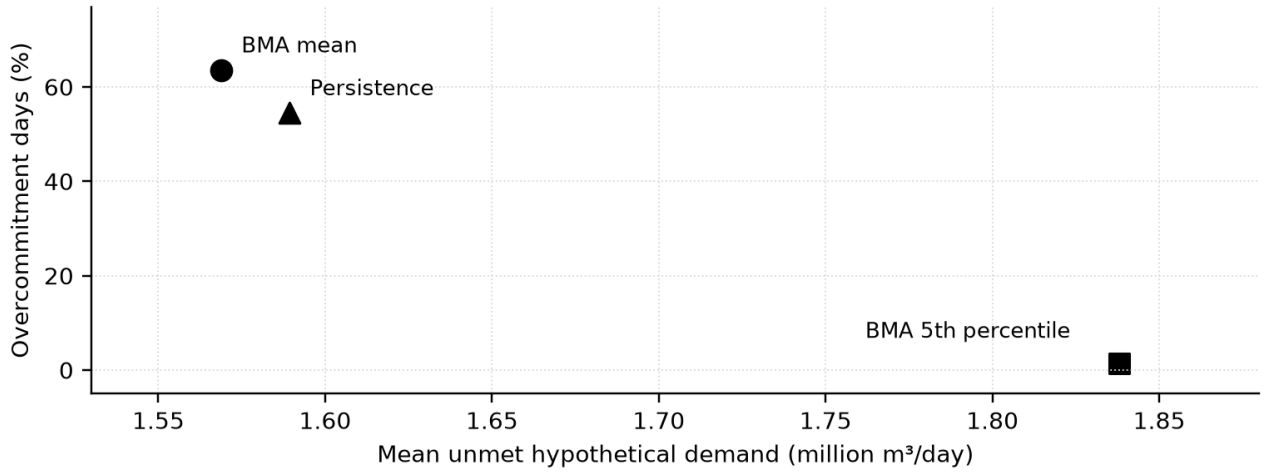


Figure 2. Lower-tail planning reduces overcommitment while increasing unmet hypothetical demand. Both axes represent undesirable outcomes; the figure shows a tradeoff, not a universal winner.

How outcomes are calculated

The plan uses forecast flow. Observed discharge is used afterward to calculate the feasible scenario budget, delivered volume, overcommitment and unmet demand. Overcommitment is planned volume above that observed budget; it does not imply actual abstraction. Foregone feasible delivery compares delivery with a perfect-information budget under the same scenario rules.

Mean foregone feasible delivery was 62,678 m³/day for persistence, 42,169 for BMA mean, and 311,501 for BMA 5th-percentile planning. The conservative policy leaves more potentially deliverable water unused. These conditional results illustrate the need to evaluate decision consequences as well as forecasting error.

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Implementation, outputs and verification

Repository area	Contents and role
src/hydrology/	data.py: ingestion/windows; models.py and train.py: forecasts; bma.py: calibrated mixture; remote_sensing.py: observation kernels; satellite_pipeline.py: CSV workflows; twin.py: state/allocation/bucket; decision_replay.py: scenario evaluation.
earth_engine/	export_observations.js and event_screening.js provide acquisition/export templates for crop, wetness and event layers.
data/	Archived USGS response, Webb supplements and source/checksum manifest. Processed series are regenerated locally; unavailable datasets are recorded explicitly.
results/	Baseline metrics, run configuration, calibration/test predictions, training history, BMA parameters and hydrograph; daily decision replay, summary JSON, tradeoff figure and validation record.
docs/ and paper/	Dataset audit, algorithm notes, first-experiment report, digital-twin protocol, and a working LaTeX manuscript with exploratory findings and outstanding validation identified.
tests/ and output/	Forecast, satellite and twin tests; existing Word progress report and PDF describing models, datasets and progress. Local checkpoints and temporary render files are excluded from Git.

Validation completed

The current project test suite passed **28 tests** on 27 October 2026. Coverage includes causal windows and train-only scaling; missing dates and source flags; model gradients; mixture means/intervals; satellite quality masks and edge fitting; spatial leakage; event screening; CSV pipelines; observation availability; demand units; allocation budgets; abstention; and water balance. Both Earth Engine scripts passed JavaScript syntax checks.

Synthetic satellite fixtures test software behavior only. The recorded CLI smoke checks exercised rice and moisture processing with synthetic inputs; they do not establish empirical map accuracy. Earth Engine execution and independent field validation remain outstanding.

Reproducibility

The recorded environment uses Python 3.12.14, PyTorch 2.14.1, NumPy 2.5.3, pandas 3.0.6, SciPy 1.18.1 and scikit-learn 1.9.1. Separate dependency locks cover the base and satellite environments. SHA-256 manifests preserve input identity, and saved predictions permit metric audits. Result-producing commands reject existing/nonempty output destinations as documented.

Install the base and satellite lock files, install the package, then run:

```
python -m hydrology.data
python -m pytest -q
python -m hydrology.train --output results/new_run
python -m hydrology.decision_replay --output results/new_replay
```

Satellite CSV commands use hydrology.satellite_pipeline with mode rice or moisture, an input file and a new output directory. See the repository README for full environment setup.

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Limitations and next steps

What the present evidence establishes

The project establishes a working forecasting benchmark, calibrated predictive mixture, tested observation/decision modules and an allocation replay. Operational digital-twin performance and the benefit of satellite-informed allocation remain unvalidated.

Remaining issue	Required next step
Forecast generalization	Use multiple seeds, longer lead times, additional catchments and nested rolling-origin evaluation. Add issue-time-available weather predictors and matched-budget comparisons.
Forecast uncertainty	Evaluate flood/low-flow regimes, estimated-record sensitivity, proper probabilistic scores and flow-dependent calibration. Reserve new untouched data for future tuning claims.
Study area and crop data	Choose and verify a basin boundary with suitable paddy coverage, discharge records and independent field labels. Evaluate buffered spatial blocks, seasons and catchments.
Wetness and turbidity	Collect footprint-, depth- and time-matched moisture measurements, and paired reflectance/NTU or suspended-solids data. Validate site-specific conversions separately.
Flood / landslide layers	Obtain independent dated event inventories and negative controls; assess omissions, false positives, radar shadow and terrain effects.
Decision value	Compare crop and moisture ablations, flow-only planning, latency and missing-data scenarios under identical demands and constraints. Calibrate crop demand and observation errors.
Operational realism	Establish actual release/vintage timing, governance rules and abstraction constraints; add routing, reservoirs or groundwater only when the intended study requires them.

Proposed evaluation sequence

Finalize the case study and reference data, execute and audit Earth Engine exports, validate each observation module, and calibrate inputs on training data. Run controlled decision experiments with measured availability, followed by 3/7/14/30-day delays and structured data gaps.

Report shortage, overcommitment, reserve violations, abstention and unit-level outcomes. Use paired hydrological block-bootstrap intervals and prespecified seeds; adjacent pixels and consecutive days are not independent replications.

Research contribution and submission readiness

The intended contribution is an evaluation of observation availability, data quality and decision consequences. Combining established algorithms alone does not establish novelty. The working manuscript needs a fuller literature comparison and independent experiments before publication or operational claims.

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Evidence and selected references

This report summarizes the project snapshot pushed on 27 October 2026. Numerical findings come from the repository's saved artifacts; models were not retrained for this document. External sources below are those recorded in the project documentation, rather than a new literature review.

Project evidence register

Evidence	Location in the repository
Scope, methods and dataset audit	README.md; docs/algorithms.md; docs/datasets.md; docs/first_experiment.md; docs/digital_twin.md
Flow data provenance	data/manifest.json and archived USGS daily-value response
Forecast results	results/baseline_seed42/metrics.json, run.json, bma.json, test_predictions.csv and calibration_predictions.csv
Decision results and verification	results/decision_replay_20261027/report.json, daily_decisions.csv and implementation_validation.json
Working manuscript	paper/river_basin_digital_twin/main.tex
Source revision	bcf55f826562cdbf8ffeab211934a12f8925f3bb

Selected research and official sources

[1] U.S. Geological Survey. Potomac River monitoring station USGS-01646500. Archived daily values retrieved 8 October 2026. Source.

[2] Webb et al. (2025). Phosphorus Removal in Constructed Treatment Wetlands: A Systematic Review. Water, 17(22), 3301. Source.

[3] Hochreiter and Schmidhuber (1997). Long Short-Term Memory. Neural Computation. Source.

[4] Vaswani et al. (2017). Attention Is All You Need. Source.

[5] Raftery et al. (2005). Using Bayesian Model Averaging to Calibrate Forecast Ensembles. Monthly Weather Review, 133. Source.

[6] Yang et al. (2024). Digital twinning of river basins towards full-scale, sustainable and equitable water management and disaster mitigation. Source.

[7] Google Earth Engine. Sentinel-1 preprocessing; MOD13A2.061 and MOD11A2.061 product documentation. Source.

[8] A Comparison of Three Trapezoid Models Using Optical and Thermal Satellite Imagery for Water Table Depth Monitoring in Estonian Bogs (2020). Remote Sensing, 12, 1980. Source.

[9] Sentinel-1 SAR Amplitude Imagery for Rapid Landslide Detection (2019). Remote Sensing, 11, 760. Source.

Further source links, including wetland access records, flood-mapping guidance and turbidity research, are retained in docs/datasets.md and docs/digital_twin.md.

Project repository: github.com/doraem-on/Hydrology_Institute-project